A Taxonomic Survey of Model Extraction Attacks

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2023 IEEE International Conference on Cyber Security and Resilience (CSR) | 979-8-3503-1170-9/23/$31.00 ©2023 IEEE | DOI: 10.1109/CSR57506.2023.10224959

***Abstract*—A model extraction attack aims to clone a machine learning target model deployed in the cloud solely by querying the target in a black-box manner. Once a clone is obtained it is possible to launch further attacks with the aid of the local model. In this survey, we analyze existing approaches and present a taxonomic overview of this field based on several important aspects that affect attack efficiency and performance. We present both early works and recently explored directions. We conclude with an analysis of future directions based on recent developments in machine learning methodology.**

I. Introduction

The success of machine learning (ML) models resulted in their widespread use in various disciplines. However, supervised training of a high-accuracy model is costly, requiring an excessive number of manually labeled data points, powerful computing resources, expert knowledge, time, and financial support. To bypass these efforts, machine learning as a service (MLaaS) platforms have emerged. These platforms serve their prediction models hosted on the cloud via a prediction API(Application Programming Interface) to the customers on a pay-per-query basis. These models are confidential and proprietary. Therefore customers only have black-box access to the model. However, this does not prevent attacks [1], [2], [3], and MLaaS services bring some novel security concerns. An adversary might intend to construct a local copy of the ML model. Such an attempt by an adversary is called a model extraction attack (MEA).

A model extraction attack may have two aims. An attacker can copy the functionality of the original model, therefore damaging the business value of the model owner. Or, the attacker can conduct further attacks, such as evasion or membership inference attacks, utilizing generated adversarial samples through the white-box substitute model. Thus, an MEA can be one of the most dangerous attacks, either by itself or as a stepping stone to further attacks. In order to facilitate a better understanding of these issues, we present a taxonomy of current research on MEAs and survey current MEA approaches in a systematic way.

To date, only Gong *et al.*[4] presents a survey on this topic. To the best of our knowledge, this is the first paper to propose a taxonomic survey covering a broad spectrum of applications specific to model extraction attacks. Our survey also covers query sample generation and attacks on generative models. Based on our taxonomic analysis, we identify and discuss several future directions.

II. MEA: Definition and Terminology

Model extraction attack refers to the attempt of an attacker to reconstruct a copy of the original ML model with less effort than training one from scratch. The primary objective of an adversary is to replicate the decision boundary or functionality of a currently trained machine learning model. Early literature is based on the equation solving approach, and recent work concentrates on the retraining technique that we briefly summarize below.

Equation-solving approaches are based on the initial work of Lowd and Meek [5] that aims to generate a sample that evades detection of a target spam filter. They obtain the parameters of a linear model to find an adversarial example rather than stealing the model exactly. This work is in the early stage of an MEA. Later, Tramèr *et al.* [1] extended this method to cover both linear and non-linear models. However, this approach is effective if the model outputs a confidence score. Thus, they proposed a retraining method for the case of membership query learning in which the model outputs only class labels.

In the retraining method, firstly, the attacker queries a group of inputs to the target model and obtains its output predictions. Then, obtained input-output prediction pairs are used to train a local copy of the target model, which is called the substitute model. The MEA is conducted in a black-box manner, meaning model parameters such as model architecture and training data distribution are unknown to the adversary. Interaction between the prediction API and the adversary is based solely on query inputs. Fig. 1 illustrates the model extraction attack methodology.

Below, we present the terminology regarding MEAs.

- The **target model** is the model targeted by the adversary. For discriminative models, the target model f_T accepts an input $x \in \mathbb{R}^n$ and produces an output $f_T(x) = y \in \mathbb{R}^K$. The target model may also be a generative model that learns the distribution of the training data and generates an image as close as possible to the learned images. Thus, the adversary aims to copy the learned distribution of the training dataset. Some existing works refer to the target model as the victim or secret model.

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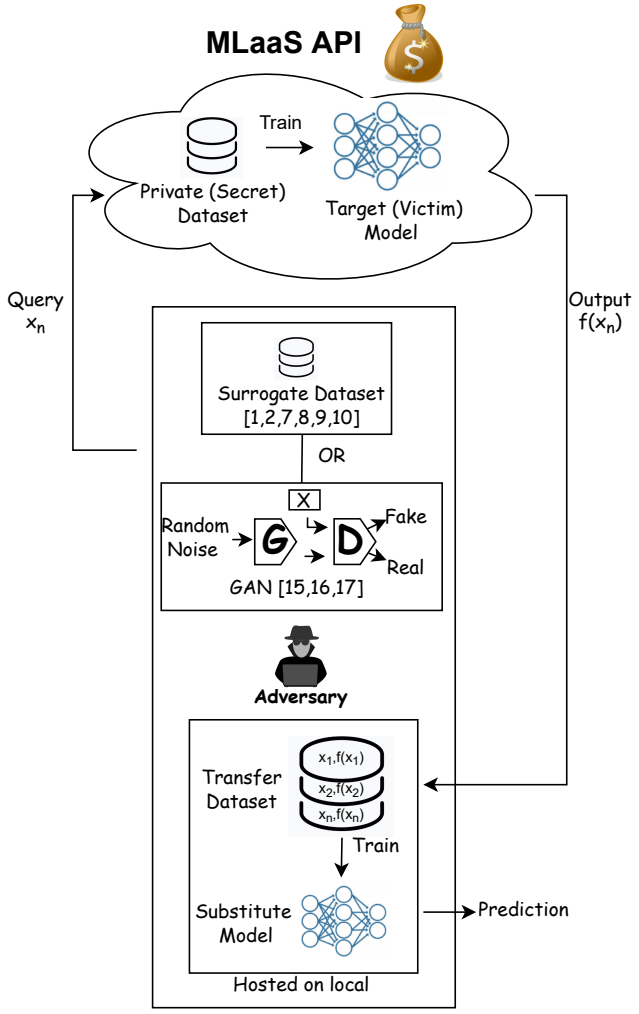


Fig. 1. Model Extraction Attack Overview

- The **substitute model** is a clone or copy of the target model. The adversary aims to construct a substitute that is functionally close to the target. Substitute model f_S is trained on a transfer dataset $D_A = \{x_i, f_T(x_i)\}_{i=0}^{N}$, which is constituted through querying the target model as an oracle. Here N defines the query budget. This model is also called knockoff, surrogate, clone, or extracted model.
- The **private dataset** $D_P = \{x_i, y_i\}_{i=0}^{M}$ refers to the target model's training dataset. This dataset is proprietary and confidential. Therefore, the adversary usually does not have access to this or its distribution. The studies also use the term secret dataset for the private dataset.
- The **surrogate dataset** is the dataset sampled by the attacker and used in the queries. It should have a distribution that is close to that of the private dataset. Studies call this dataset a thief, adversary, or surrogate dataset.
- For **accuracy extraction**, the purpose is to maximize task accuracy of the substitute model $max[argmax(f_s(xi)) = y_i]$.
- For **fidelity extraction**, an adversary aims to increase the agreement between target and substitute model $max[argmax(f_t(x_i)) = argmax(f_s(x_i))]$. In this strategy, the substitute model's misclassification result matches the target model's output.

III. Attack Taxonomy and Overview

The aim of presenting a taxonomy is to clarify the domain for both owners and defenders. If we know the attack surface in all directions, we can focus on gaps to mitigate vulnerabilities and produce defense techniques for protection or detection. Aware of this importance, we have elaborated the taxonomy by considering domain papers broadly.

Fig. 2 illustrates the proposed MEA taxonomy. We first divide the MEAs into two based on the main method used, either the equation solving approach of Lowd and Meek [5] or the retaining method that most current approaches rely on.

There are limited studies on the equation-solving method due to its limitations, such as requiring confidence values as output. However, it is easier to repetitively query the target model to obtain class labels, which is sufficient to retrain. The nature of generative and discriminative models and related retraining strategies are different. Therefore, we categorize the retraining approaches according to the type of target model.

An attacker repetitively queries the target model with samples selected from a substitute dataset. Consequently, query distribution depends on the selected substitute dataset source which can be a *natural dataset* or a *sythetic dataset*. While the sampling technique is essential for selecting query samples among natural datasets, generation techniques differ in the construction of synthetic datasets. Accordingly, we divide the natural dataset into groups according to query sampling techniques that aim to reduce the query budget: *Active learning*, *semi-supervised learning*, *reinforcement learning*, and *random sampling*. Similarly, we divide approaches that rely on a synthetic substitute dataset based on how they craft synthetic samples: *generative sampling*, *model inversion*, *jacobian-based*, and *feature-based optimization* algorithms.

IV. MEA by Equation Solving

Lowd and Meek [5] first proposed the equation-solving attack to investigate the adversarial attack on linear classifiers, such as naive Bayes, and the maximum entropy model. They aimed to conduct an evasion attack by finding an adversarial sample, which a target spam filter misclassifies, rather than learning the actual decision boundary of the classifier. They assume that an adversary can access the cost function and obtain one positive (spam mail) and one negative (non-spam mail) sample. In this way, they reconstruct the decision boundary, thus the weights of the classifier, by using a binary search between a positive and negative sample point with a certain number of queries. However, the proposed algorithms are not query-efficient requiring a polynomial number of queries.

Later, Tramèr *et al.* [1] solved output equations of linear models, which have d parameters, in $d + 1$ queries. For non-linear models, they used optimization techniques such as stochastic gradient descent. Strongly convex loss functions

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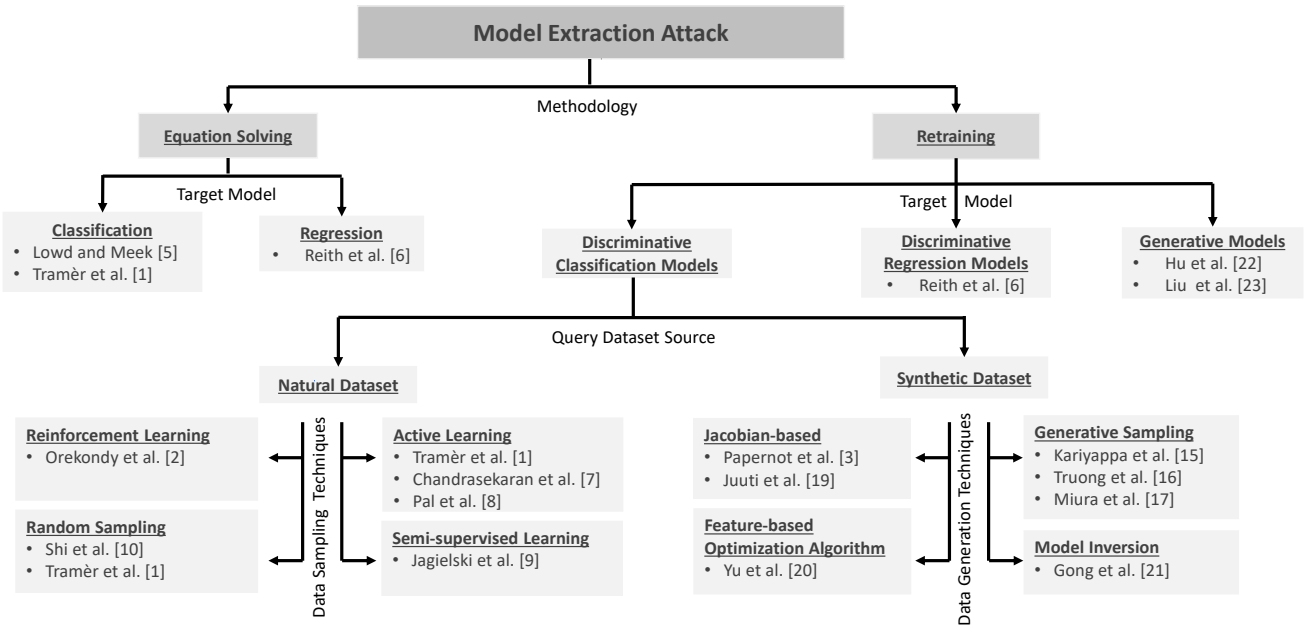


Fig. 2. Model Extraction Attack Taxonomy

converge to a global minimum. Thereby functionality of f_T and f_S matches precisely. However, for not strongly convex functions (e.g., multilayer perceptron (MLP)), the behavior of f_S approximates the behavior of f_T. Note that, the equation solving attack targets models that output confidence values.

Reith *et al.* [6] is the first and only study that conducts a white-box extraction attack on a regression model, which is quadratic kernel Support Vector Regression Machines (SVRs). To find the weights of an SVR, they query the target model with a vector x whose value is 1 for the specific position of a corresponding weight and 0 for others. They repeat this querying process until they obtain all the weights and extract the bias values by querying the oracle with a zero vector x.

V. MEA by Retraining

Tramèr *et al.* [1] first proposed the retraining approach for targeting models that only give the class labels instead of probabilities. They obtain input-output pairs (x,f_T(x)) by repetitively querying the target model, and train a local substitute model with a similar decision boundary. Therefore, this process is called the retraining approach.

This approach can target both generative and discriminative models. Generally, among discriminative models, attacks are carried out on classification models. However, the only study that performs extraction of the regression model by retraining is Reith *et al.* [6] who extended Tramèr *et al.*'s retraining strategy for SVRs with arbitrary kernels.

A. Retraining Discriminative Classification Models

A retraining attack relies on obtaining the input-output pairs (x, f_T(x)) by querying the oracle f_T by choosing samples from natural or synthetic datasets. We first discuss approaches that use a natural surrogate dataset using one of several published datasets, such as ImageNet [11], CIFAR-10 [12], German Traffic Sign Recognition Benchmark (GTSRB) [13], Modified National Institute of Standards and Technology database (MNIST) [14] etc. Then, we discuss approaches that synthetically generate substitute dataset samples.

1) Query Sampling from Natural Datasets: The query sampling strategy is the second significant point affecting attack efficiency after the query source. Samples located close to the decision boundary reveal more information. Thus, an adversary's purpose is to find such samples to extract the model with a limited number of queries. Table I lists approaches that sample queries from a natural surrogate dataset.

The simplest query sampling approach is to just select samples uniformly at random. Shi *et al.* [10] mounted an MEA on naive Bayes, Support Vector Machine (SVM), and Deep Neural Network (DNN) models in the text classification domain by randomly selecting the queries among private datasets. As discussed above, [1], [2] also used random uniform sampling to evaluate their studies.

Instead of pure random selection, Chandrasekaran *et al.* [7] used the extended adaptive training (EAT) approach to decrease the query budget sharply (5x-224x) compared to [1]. The EAT approach chooses one sample with the minimum confidence score $|y_i(x_i)|$ among randomly selected k random inputs $x_1...x_k$ for labeling in each iteration.

One way to sample queries is to first select an initial subset at random and label this set by querying f_T. We can then train an initial substitute model, f_S, on this labeled data, which we can exploit in sampling the queries using an active learning approach. Tramèr *et al.* [1] proposed an adaptive retraining approach that select samples that f_S is least certain about. Similarly, Pal *et al.* [8] trains an initial f_S and labels the remaining parts of the surrogate dataset with it. They

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TABLE I
THE LITERATURE WHICH USES NATURAL DATA QUERIES

Reference	Aim	Target Model	Surrogate Dataset	Private Dataset	Data Type	Query Budget	Target MLaaS
Tramèr et al. [1]	CLONE	LR/MLR/NN k-SVM/DT	Digits/German Credit Adult/Steak Survey	Digits/German Credit Adult/Steak Survey	Image	650/1150 1485/4013	Amazon BigML
Chandrasekaran et al. [7]	CLONE	k-SVM/DT		Breast Cancer/Adult Mushroom/Diabetes	Image	119/48 1001/166	
Pal et al. [8]	ADV	CNN	ILSVRC	MNIST/CIFAR10 GTSRB	Image	30k	
Jagielski et al. [9]	CLONE	CNN	ImageNet/CIFAR10 SVHN	Social Media CIFAR10/SVHN	Image	4k	
Orekondy et al. [2]	CLONE	CNN	ILSVRC OpenImages	Caltech/CUBS200 Indoor/Diabetic	Image	60k	Facial Characteristics
Shi et al. [10]	CLONE	SVM/NB	Reuters-21578	Reuters-21578	Text	859	

LR: Logistic Regression, MLR: Multilayer Perceptron, CNN: Convolutional Neural Network, DNN: Deep Neural Network, k-SVM: Kernel-Support Vector Machine, DT: Decision Tree, NB: Naive Bayes. 'Aim': The adversary's aim of functionality stealing ('clone') or adversarial sample generation ('adv').

choose the subsequent query samples using pool-based active learning based on these labels from f_S. Then the target model is queried for label refinement. f_S is retrained on refined labels and this process is repeated until the query budget is spent. Furthermore, they generated a highly transferable adversarial sample by exploiting the trained substitute model f_S. Orekondy *et al.* [2] uses reinforcement learning to select the samples. A learning policy is updated at each sampling time according to a predefined reward function.

Alternatively, the adversary may only query a subset of the surrogate dataset and use semi-supervised learning to take advantage of the remaining unlabeled part. Jagielski *et al.* [9] exploited two semi-supervised learning methods for accuracy extraction; rotation loss and MixMatch. They showed that only querying 250 samples and using the semi-supervised approach on the remaining unlabeled data improves upon labeling the whole dataset when the dataset size is small enough.

2) Query Sampling by Synthetic Sample Generation: For some domains, such as medical diagnosis, it is challenging to find samples to query. Consequently, several studies proposed constructing query samples using synthetic data generation techniques. Table II shows the literature that mounts an MEA using synthetically built queries.

Data-free model extraction attack, in which a private dataset is unknown to the attacker/user, is recently proposed ([16], [15], [17]). In this approach, the adversary trains a generative model to generate samples instead of collecting natural ones. Kariyappa *et al.* [15] is the first to propose a data-free MEA that uses a generative adversarial network (GAN) generator. The generator training requires computation of the gradient of the loss with respect to target model parameters. However, since this is a black-box attack, gradients are approximated using the forward differences zeroth-order gradient approximation method. It computes directional derivatives of a function f at point x along m random directions. The drawback of this is the increased query budget. Similarly, Truong *et al.* [16] investigate the effect of different loss functions on GAN performance. Since KL-divergence loss suffers from the vanishing gradient problem while the substitute model's results converge to the target model's probabilities, they instead used ℓ_1 norm loss between the substitute and target model logits (pre-softmax results). Both studies obtained similar substitute accuracy on the same dataset for an equal number of queries.

Unfortunately, these two studies show that a data-free model extraction attack needs more queries than a surrogate dataset since the adversary has to train a generative model. To decrease the number of queries, Miura *et al.* [17] utilized a gradient-based explanation method, the *Vanilla Gradient*, to perform an MEA on gradient-based explainable AI. This method shows the effect of each pixel of the input image on the confidence scores by calculating the gradient of prediction results with respect to input pixels which is the part that is approximated in the studies [16],[15]. In this way, they aim to remove the gradient approximation burden on the number of queries. However, they get nearly the same experimental results as Truong *et al.* for the same experimental setup.

Tasumi *et al.* [18] proposed a novel MEA called `TEMPEST` on tabular data which is common in domains such as healthcare, finance, and transportation. They argue that using publicly available statistical information while generating query samples performs an efficient MEA. This approach presents comparable results to the state-of-the-art when the statistical information is derived from one part of the target model's dataset. However, it results in a low-accuracy substitute model when statistics are obtained from a public dataset.

Instead of training a GAN, Gong *et al.* [19] were inspired by model inversion techniques. The inversion process is considered an encoder-decoder architecture in which the target model encodes the input in a prediction vector. The inversion model decodes it to obtain a new sample. First, they train an initial substitute model selecting query samples by the core-set algorithm. Then, they choose the high-confidence samples among previously chosen query datasets to take their inversion. In this manner, they obtain a single representative inverted sample for each target model class. Finally, the substitute model is retrained on inverted samples. They achieved higher

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TABLE II
THE LITERATURE THAT GENERATES SYNTHETIC SAMPLES

Reference	Aim	Target Model	Surrogate Dataset	Private Dataset	Data Type	Query Budget	Target MLaaS
Kariyappa et al. [15]	CLONE	CNN		GTSRB/CIFAR-10 SVHN/FashionMNIST	Image	30M	
Truong et al. [16]	CLONE	CNN		CIFAR-10/SVHN	Image	20M/2M	
Miura et al. [17]	CLONE	CNN		CIFAR-10/SVHN	Image	20M/2M	
Tasumi et al. [18]	CLONE	MLP		Adult/Cancer Diabetes/Arrhythmia	Tabular Data	2500/2500 2500/9500	
Papernot et al. [3]	ADV	DNN/LR SVM	MNIST GTSRB	MNIST GTSRB	Image	2k	MetaMind Google/Amazon
Gong et al. [19]	CLONE	CNN	EMNIST Letters ImageNet	MNIST/CIFAR-10 GTSRB	Image	30k	Alibaba Cloud
Juuti et al. [20]	ADV	CNN	MNIST/GTSRB	MNIST/GTSRB	Image	100k	
Yu et al. [21]	CLONE	CNN		GTSRB VGG Flowers	Image	$\sim$3k	Clarifai NSFW MS Cloud Vision Google AutoML Face++ Emotion

CNN: Convolutional Neural Network, MLP: Multilayer perceptron, DNN: Deep Neural Network, LR: Logistic Regression, SVM: Support Vector Machine. 'Aim': The adversary's aim of functionality stealing ('clone') or adversarial sample generation ('adv').

fidelity and accuracy results than state-of-the-art models with a lower query budget. Additionally, they showed that a well-known defense strategy, PRADA [20], could not detect their extraction attack since the generated query distribution was close to normal. Papernot *et al.* [3] proposed the jacobian-based dataset augmentation (JBDA) approach to generate a synthetic dataset. They assume an adversary can access a limited number of samples in the target model task. After selecting initial query samples, they query the target model for labeling, and these samples are augmented with JBDA. The JBDA is based on identifying the varying directions of the substitute model output. The sign of the Jacobian matrix of an input x for the corresponding label is evaluated and added to the original input to get a new synthetic sample. In each iteration samples are selected using reservoir sampling, thereby limiting the exponential growth of querying. After obtaining a white-box substitute model, they take this extraction attack a step further by crafting an adversarial sample that is misclassified by both the substitute and target models.

Juuti *et al.* [20] also aimed to craft high transferability adversarial samples. They used a limited number of seed samples from the surrogate or the original train datasets. In the first training round, natural samples are used, while the subsequent rounds utilize synthetic samples in substitute model training. They explore the impact of different sample crafting strategies on the MEA performance and transferability rate of the adversarial samples.

Yu *et al.* [21] exploited transfer learning to get high-accuracy substitute models by retraining the last layers of pre-trained models. The target model labels a hand-crafted dataset that is used in the fine-tuning process. They examined five different adversarial sample crafting strategies to generate the most informative ones. Seed samples of a synthetic dataset are chosen among unlabeled datasets via active learning.

B. Retraining Generative Models

Mounting an MEA on generative models differs from discriminative models. Firstly, generative models return images or data vectors as output instead of class probabilities or labels. Secondly, GANs use different metrics for accuracy evaluation since the adversary aims to learn the data distribution. Accuracy extraction aims to minimize the difference in data distribution between the target and substitute models. Fidelity extraction aims to maximize the similarity of data distribution between the substitute and the target model's training datasets.

Hu *et al.* [22] conducted an extraction attack on Progressive GAN (PGGAN) and Spectral Normalization GAN (SNGAN) arhitectures, and proposed a defense technique for this attack. The accuracy of the extraction attack was measured in a black-box manner by querying the generator of the target GAN. This way, they obtain a training set to retrain the substitute GAN model. However, fidelity extraction is conducted in a partial black-box or white-box attack manner. They assumed the adversary had partial real data (for a partial black-box attack) or the target model's discriminator (for a white-box attack). In addition, they applied a case study including transfer learning to present the transfer of the substitute model to another domain. Liu *et al.* [23] investigated the extraction of PixelRNN/PixelCNN, variational autoencoder (VAE), and GAN, but they used the 1-Nearest Neighbor classifier as the evaluation metric of attack to evaluate the distribution similarity of the victim and the target model.

VI. DISCUSSION AND FUTURE DIRECTIONS

Query budget is one of the most important factors that affect the attack cost and the ability to detect an attack. In order to decrease this budget, current works rely on techniques such as reservoir sampling [3], reinforcement learning [2], core-set algorithm [19] to sample a subset of the

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queries or semi-supervised learning [9] to combine queries with unlabeled data. An alternative to sampling a natural dataset is to create synthetic examples either by training a generative network [16], [15], [17] or active data augmentation schemes [19], [20], [3] exploiting a partially trained model.

Another possible direction is to increase query efficiency by exploiting advanced training approaches such as curriculum learning [24] and contrastive learning [25]. Our survey reveals that, currently, this is a relatively under-explored direction.

Recently developed large language models [26] and diffusion based text-to-image generation approaches [27] have impressive generative capabilities. Instead of training a GAN from scratch, exploiting these models might also be feasible in mounting an MEA. However, how to generate the input to these networks in an active learning scheme or whether the generated data will be of sufficient fidelity are open questions.

Substitute architecture selection is another important aspect. [3], [2] shows the attack feasibility on complex models and current studies show that substitute models which are from the same family or more complex than the target model yield higher MEA performance.

Obtaining a highly accurate substitute model is one of the aims, but adversarial sample creation that fools the target model can be the purpose of the adversary. Most of the current works aim to clone the target model, but fewer works such as [3], [20] generate highly transferable adversarial samples.

Our taxonomy also reveals that MEAs are heavily studied in classification models, which are more commonly used in MLaaS platforms. However, we could not encounter any retraining study that steals a regression model. Potentially, this is a more challenging problem since a regression model requires continuous target values instead of discrete labels.

VII. Conclusion

We have presented a taxonomic survey of MEAs on various ML models. Our survey covers both early and recent works, those relying on query sampling or synthetic generation. It also covers recent directions such as stealing generative networks. Analyzing the current works in light of the proposed taxonomy, we have discussed future directions that might increase MEA efficiency and expand to new architectures.

Acknowledgment

This work was supported in part by The Scientific and Technological Research Council of Turkey (TUBITAK) through the 1515 Frontier Research and Development Laboratories Support Program under Project 5169902.

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